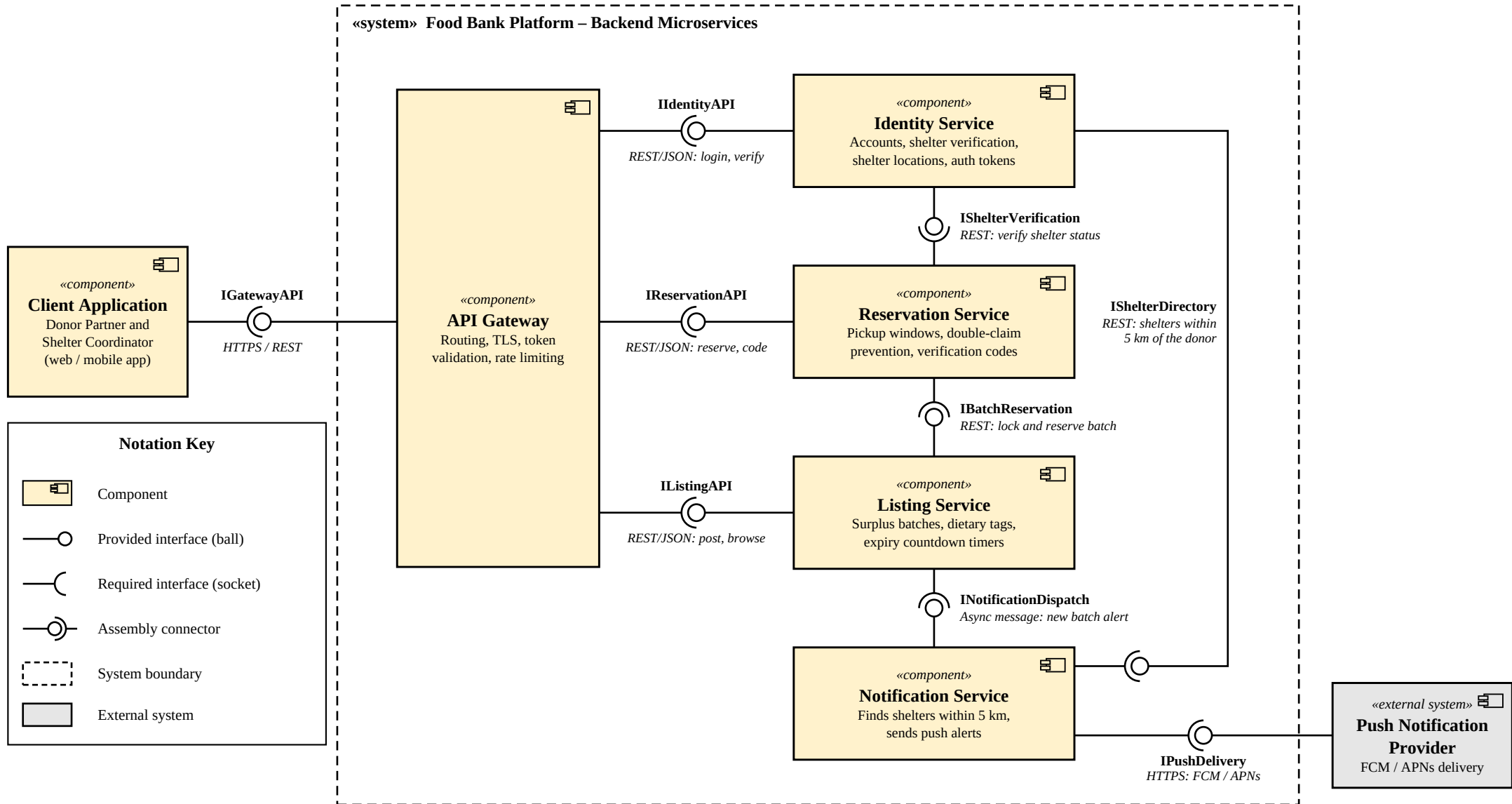


Lab 3: UML Component Diagram

Food Bank Surplus Redistribution Platform (Problem Statement #62) – Microservices Architecture



Reading the diagram: every assembly connector joins the component that provides an interface (ball) to the component that requires it (socket). Requests travel from the socket side to the ball side and responses return the same way.

Data ownership: each microservice owns a private datastore (database-per-service). No service reads another service's data directly; it must call that service's provided interface.

Layout: customer-facing components are on the left, the API Gateway is the single entry point to the backend, and the four domain microservices and the external push provider are on the right.

Component and Interface Catalogue

Supporting tables for the component diagram

A. Components

#	Component	Responsibilities
1	Client Application (web / mobile)	Used by the Donor Partner and the Shelter Coordinator to post surplus batches, browse the claim board, reserve pickup windows and receive push alerts.
2	API Gateway	Single entry point to the backend: routes requests to services, terminates TLS, validates authentication tokens and rate-limits clients.
3	Identity Service	Registers donors, verifies and stores shelter accounts and locations, and issues authentication tokens.
4	Listing Service	Creates surplus food batches with quantity, dietary tags and an expiry countdown timer; removes expired batches from the claim board.
5	Reservation Service	Manages pickup windows, prevents two shelters from claiming the same batch, and generates the pickup verification code.
6	Notification Service	Finds verified shelters within 5 km of a donor and sends push alerts when high-quantity perishable food is listed.
7	Push Notification Provider (external)	Third-party service (FCM / APNs) that delivers push messages to shelter devices.

B. Interfaces

#	Interface	Provided by	Required by	Protocol	Data flow (request and response)
1	IGatewayAPI	API Gateway	Client Application	HTTPS / REST	Login, post, browse and reserve requests; JSON responses including the pickup verification code.
2	IIdentityAPI	Identity Service	API Gateway	REST / JSON	Credentials or shelter ID; returns auth token and verification status.
3	IListingAPI	Listing Service	API Gateway	REST / JSON	Batch details (quantity, dietary tags, expiry); returns active batches and countdown timers.
4	IReservationAPI	Reservation Service	API Gateway	REST / JSON	Batch ID, shelter ID and pickup window; returns reservation status and verification code.
5	IShelterVerification	Identity Service	Reservation Service	REST	Shelter ID; returns verified or not verified.
6	IBatchReservation	Listing Service	Reservation Service	REST	Batch ID; locks the batch if it is still active and unexpired, returns confirmation or rejection.
7	INotificationDispatch	Notification Service	Listing Service	Async message	New high-quantity batch event: batch ID, quantity and donor location.
8	IShelterDirectory	Identity Service	Notification Service	REST	Donor location and 5 km radius; returns verified shelters and their device tokens.
9	IPushDelivery	Push Notification Provider	Notification Service	HTTPS (FCM / APNs)	Push payload for shelter device tokens; returns delivery status.